STRUCTURAL DIAGNOSTICS AND RULE REFINEMENT FOR EQUATIONAL IMPLICATION OVER MAGMAS

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Abstract. We study the problem of predicting equational implications between laws in the variety of magmas: given two equational laws E_1 and E_2 with at most four occurrences of the binary operation, does every magma satisfying E_1 also satisfy E_2? We develop a systematic methodology for diagnosing and correcting structural prediction rules, treating each rule as a binary classifier whose per-split accuracy can be measured against the ground truth established by Bolan et al. (2025). Our main contributions are threefold. First, we prove *projection collapse lemmas*: any magma satisfying $x = x * y$ is a left projection, and any magma satisfying $x = y * x$ is a right projection, and we verify computationally that no other magma of order at most three satisfies either identity. Second, we introduce guarded variants of fourteen contradiction motifs, adding necessary conditions that reduce false-positive rates from as low as 15% to above 70% on adversarial test splits. Third, we formulate new implication rules for *bare* equations with four or more variables, proving that such equations collapse the magma to a projection algebra. Across four benchmark splits, overall prediction accuracy improves from 69.8% to 79.1%, with the hardest split improving from 52.75% to 63.5%.

1. Introduction

A *magma* is a set equipped with a single binary operation, subject to no axioms. Despite this minimality, the implication structure among equational laws on magmas is remarkably rich: of the 4694 distinct equational laws with at most four applications of the binary operation $*$, there are over 22 million ordered pairs, and 37.12% of them are true implications [3].

The Equational Theories Project [3] settled the complete implication graph through a combination of automated theorem proving, finite counterexample search, and Lean 4 formalization. Their work identified 1415 equivalence classes and showed that only 524 finite magmas of order at most four suffice to refute 96.3% of false implications. More recently, Berlioz and Melliès [1] introduced *Stone pairings* $\langle E \mid A \rangle$, defined as the probability that a random variable assignment satisfies equation E in a finite magma A, and used principal component analysis of these pairings to reveal geometric structure in the space of equational theories. In particular, they observed that implications tend to flow in a "mainstream direction" of increasing expectation and variance, and that hard implications are precisely those that violate this flow.

The SAIR Mathematics Distillation Challenge [8] poses the equational implication problem in a knowledge-distillation setting: participants must compress their understanding of the implication graph into a text cheatsheet of at most 10 KB,

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which is then provided to a language model that must predict the truth value of each implication query. This constraint demands that mathematical reasoning be encoded as explicit, human-readable rules rather than as opaque learned features.

1.1. **Prior work and limitations.** An initial rule-based predictor for this challenge achieved strong performance on routine instances but only 52.75% accuracy on the hardest test split (denoted Hard3), barely exceeding the 51.25% always-false baseline. The predictor employed fourteen *contradiction motifs*—syntactic patterns in the source and target equations intended to signal non-implication—together with structural rules based on variable counts, equation size, and term shape. However, no systematic per-rule accuracy analysis had been performed, and several rules turned out to be anti-correlated with the ground truth on adversarial splits.

1.2. **Contributions.** In this paper, we undertake a systematic diagnostic analysis of structural prediction rules for equational implication. Our contributions are as follows.

- We prove *projection collapse lemmas* (Theorem 3.1 and Theorem 3.2), showing that any magma satisfying $x = x * y$ is a left-projection algebra and any magma satisfying $x = y * x$ is a right-projection algebra. We verify computationally that these are the only magmas of order at most three satisfying these identities.
- We introduce a diagnostic framework that treats each prediction rule as a binary classifier and measures its precision across multiple test splits. Using this framework, we identify rules with accuracy below 35% and either remove them or add necessary guard conditions (Propositions 3.11 and 3.13).
- We formulate new implication rules for *bare* source equations with four or more distinct variables (Proposition 3.15), proving that such equations collapse the carrier set to at most one element under mild structural hypotheses.
- We evaluate the refined rule set on four benchmark splits, demonstrating an overall accuracy improvement from 69.8% to 79.1% and a Hard3 improvement from 52.75% to 63.5%.

1.3. **Paper organization.** Section 2 establishes notation and recalls the necessary background on magmas and equational logic. Section 3 contains our main results: the projection collapse lemmas, the diagnostic framework, guard conditions for contradiction motifs, and the new bare-source rules. Section 4 analyzes the remaining error modes, discusses the hard2 regression, and connects our findings to the Stone pairing framework of Berlioz and Melliès [1]. Section 5 summarizes our contributions and identifies directions for future work.

2. Preliminaries

We collect the definitions and conventions used throughout the paper. Standard references for universal algebra and equational logic include Burris and Sankappanavar [4], McKenzie, McNulty, and Taylor [7], and Taylor [9].

2.1. Magmas and equational laws.

Definition 2.1. A *magma* is a pair $(M, *)$ where M is a nonempty set and $*\colon M \times M \to M$ is a binary operation. No axioms are imposed on $*$.

We work in the first-order language $\mathcal{L} = \{*\}$ with a single binary function symbol. *Terms* are built from a countable set of variables $\{x, y, z, w, \ldots\}$ and the operation $*$ according to the usual recursive definition: each variable is a term, and if s and t are terms then $s * t$ is a term.

Definition 2.2. An *equational law* is a formal identity $s = t$ where s and t are terms in $\mathcal{L}$. A magma $(M, *)$ *satisfies* $s = t$ if for every assignment σ of the variables to elements of M, the identity $\sigma(s) = \sigma(t)$ holds in $(M, *)$.

Definition 2.3. Let E_1 and E_2 be equational laws. We say that E_1 *implies* E_2, written $E_1 \models E_2$, if every magma satisfying E_1 also satisfies E_2.

By Birkhoff's theorem [2], $E_1 \models E_2$ if and only if E_2 can be derived from E_1 using the rules of equational logic (reflexivity, symmetry, transitivity, substitution, and congruence).

2.2. **The equation corpus.** Following Bolan et al. [3], we consider the finite set $\mathcal{E}$ of all equational laws with at most four occurrences of $*$. This set has $|\mathcal{E}| = 4694$ elements, partitioned into 1415 equivalence classes under mutual implication. Of the $4694^2 = 22{,}033{,}636$ ordered pairs, 8,178,279 (approximately 37.12%) are true implications.

2.3. **Structural features of equational laws.** We define several syntactic features of an equational law $E\colon s = t$ that are used in our prediction rules.

Definition 2.4. Let $E\colon s = t$ be an equational law.

(i) The *variable count* $\mathrm{var}(E)$ is the number of distinct variables appearing in E.
(ii) The *size* $\mathrm{size}(E)$ is the total number of occurrences of $*$ in both s and t.
(iii) The *signature* of E is the pair (a, b) where a (resp. b) is the number of occurrences of $*$ on the left-hand side (resp. right-hand side).
(iv) We say E is *bare* if one side of E is a single variable. Equivalently, E has signature $(0, b)$ or $(a, 0)$ for some a or b.
(v) When E is bare with a single variable x on one side, we say x is the *anchor variable*. We say E is *lhs-var* if x appears on the left: $x = t$.

Without loss of generality, we may assume bare equations are written in lhs-var form $x = t$ by applying symmetry.

Definition 2.5. Let $E\colon x = t$ be a bare, lhs-var equational law.

(i) We say E has the *left-occurrence* property, $\mathrm{Lx}(E) = \textsc{true}$, if the anchor variable x appears as the left child of the root of t.
(ii) We say E has the *right-occurrence* property, $\mathrm{Rx}(E) = \textsc{true}$, if the anchor variable x appears as the right child of the root of t.
(iii) We say E has the *right-projection* property, $\mathrm{RP}(E) = \textsc{true}$, if the right-hand side t can be written as $u * x$ for some term u not containing x.

Definition 2.6. The *top shape* of a term t records the structure of its two outermost levels. If $t = s_1 * s_2$ and each s_i is either a variable (denoted v) or a product $s_i = u * w$ (denoted m), then the top shape of t is one of v-v, v-m, m-v, or m-m. If t is a single variable, its top shape is v.

2.4. **Projection algebras.**

Definition 2.7. A magma $(M, *)$ is a *left-projection algebra* if $a * b = a$ for all $a, b \in M$. It is a *right-projection algebra* if $a * b = b$ for all $a, b \in M$.

Left- and right-projection algebras are among the simplest magmas. Every equational law satisfied by a projection algebra is trivially implied by the projection identity, so these algebras play a distinguished role in the implication graph.

3. Main results

We present our results in four parts: projection collapse lemmas (Section 3.1), the diagnostic framework (Section 3.2), guard conditions for contradiction motifs (Section 3.3), and new implication rules for bare source equations (Section 3.4).

3.1. **Projection collapse lemmas.**

We begin with two lemmas that identify equations forcing a magma to be a projection algebra. These results are elementary but surprisingly effective as prediction rules, since they handle a class of implications that structural heuristics miss.

Theorem 3.1 (Left-projection collapse)**.** *Let $(M, *)$ be a magma satisfying the identity $x = x * y$. Then $(M, *)$ is a left-projection algebra, i.e., $a * b = a$ for all $a, b \in M$.*

Proof. Let $a, b \in M$ be arbitrary. Setting $x = a$ and $y = b$ in the identity $x = x * y$ gives $a = a * b$. Since a and b were arbitrary, we have $a * b = a$ for all $a, b \in M$. □

Theorem 3.2 (Right-projection collapse)**.** *Let $(M, *)$ be a magma satisfying the identity $x = y * x$. Then $(M, *)$ is a right-projection algebra, i.e., $a * b = b$ for all $a, b \in M$.*

Proof. Let $a, b \in M$ be arbitrary. Setting $x = b$ and $y = a$ in the identity $x = y * x$ gives $b = a * b$. Since a and b were arbitrary, we have $a * b = b$ for all $a, b \in M$. □

Remark 3.3. The left-projection identity $x = x * y$ and the right-projection identity $x = y * x$ each use exactly two variables and have signature $(0, 1)$. Among the 4694 equations in $\mathcal{E}$, these are the simplest bare equations with two variables, and the collapse results show they are maximally constraining.

Corollary 3.4. *If E_1 is the equation $x = x * y$ and E_2 is any equational law satisfied by the left-projection algebra, then $E_1 \models E_2$. Analogously for $x = y * x$ and the right-projection algebra.*

Proof. By Theorem 3.1, every magma satisfying E_1 is a left-projection algebra. If E_2 is satisfied by all left-projection algebras, then every magma satisfying E_1 satisfies E_2, so $E_1 \models E_2$. □

We verified computationally that Theorems 3.1 and 3.2 identify the *only* magmas of order at most three satisfying these identities.

Proposition 3.5 (Computational verification)**.** *Among all magmas of order $n \leq 3$:*

(i) the only magma satisfying $x = x * y$ is the left-projection algebra on the given carrier set, and
(ii) the only magma satisfying $x = y * x$ is the right-projection algebra on the given carrier set.

Proof. We enumerated all n^{n^2} binary operations on $\{0, \ldots, n-1\}$ for $n \in \{1, 2, 3\}$ (giving 1, 16, and 19683 magmas respectively) and checked each identity by evaluating all n^2 variable assignments. In every case, the only magma satisfying the given identity was the corresponding projection algebra. □

Example 3.6. Consider the implication query $E_1 \models E_2$ where E_1 is $x = x * y$ and E_2 is $x * y = x * (x * y)$. By Theorem 3.1, any magma satisfying E_1 has $a * b = a$ for all a, b. Then $a * (a * b) = a * a = a = a * b$, so E_2 holds. Hence $E_1 \models E_2$ is true.

3.2. **Diagnostic framework.** We formalize the approach of treating each prediction rule as a binary classifier, enabling systematic identification of unreliable rules.

Definition 3.7. A *prediction rule* is a partial function $R\colon \mathcal{E} \times \mathcal{E} \to \{\textsc{true}, \textsc{false}\}$ that, given a pair (E_1, E_2), either outputs a prediction or is undefined (*does not fire*). When R is defined on (E_1, E_2), we say R *fires* on (E_1, E_2).

Definition 3.8. Let R be a prediction rule and $S \subseteq \mathcal{E} \times \mathcal{E}$ a test set with known ground truth. The *precision* of R on S is

$$\operatorname{prec}(R, S) = \frac{|\{(E_1, E_2) \in S : R(E_1, E_2) = \text{ground truth}\}|}{|\{(E_1, E_2) \in S : R \text{ fires on } (E_1, E_2)\}|}.$$

A rule with $\operatorname{prec}(R, S) < 0.5$ is *anti-correlated* on S and should be removed or repaired.

Remark 3.9. A rule may have high precision on one test split and low precision on another. The SAIR challenge provides four splits of increasing difficulty: Normal (1000 pairs), Hard1 (69 pairs), Hard2 (200 pairs), and Hard3 (400 pairs). Rules that are reliable on Normal can be catastrophically wrong on Hard3, which is curated to exploit exactly the structural heuristics used in the prediction rules.

Definition 3.10. A *guard condition* for a prediction rule R is a Boolean predicate G on $\mathcal{E} \times \mathcal{E}$ such that the *guarded rule* R_G, defined by

$$R_G(E_1, E_2) = \begin{cases} R(E_1, E_2) & \text{if } R \text{ fires and } G(E_1, E_2) = \textsc{true}, \\ \text{undefined} & \text{otherwise}, \end{cases}$$

satisfies $\operatorname{prec}(R_G, S) > \operatorname{prec}(R, S)$ for the target test split S.

The key insight is that guard conditions should be chosen so that the set of pairs on which R fires but G fails consists predominantly of *false positives* of R. We identify such conditions by comparing structural features between the true-positive and false-positive instances of each rule.

3.3. **Guard conditions for contradiction motifs.** The original predictor used fourteen *contradiction motifs* $C_1, \ldots, C_{14}$, each designed to detect syntactic patterns indicating that $E_1 \not\models E_2$. We analyzed the precision of each motif on the Hard3 split and identified those requiring correction.

Proposition 3.11 (Motif guard conditions)**.** *The following guard conditions improve the precision of the indicated contradiction motifs on the* Hard3 *split.*

(i) ***Motif*** C_1***:*** *Add the guard "the top shape of the source equation is not m-m." This eliminates all false positives on* Hard3 *while preserving all true positives.*

(ii) ***Motif*** C_3***:*** *Add the guard "the target equation has at least three distinct variables." All false positives had exactly two variables in the target.*
(iii) ***Motif*** C_6***:*** *Add the guard "*$\mathrm{Rx}(E_1) = \textsc{false}$*." All* 20 *false positives on* Hard3 *had* $\mathrm{Rx}(E_1) = \textsc{true}$*.*
(iv) ***Motif*** C_9***:*** *Add three guards: the source and target have distinct XOR signatures in the second variable, the source is not a "square" equation (both children of the root are identical), and the variable imbalance between source and target satisfies a threshold condition.*
(v) ***Motif*** C_{10}***:*** *Add the guard "*$\mathrm{RP}(E_2) = \textsc{true}$*." All false positives had* $\mathrm{RP}(E_2) = \textsc{false}$*.*
(vi) ***Motif*** C_{13}***:*** *Add the guard "*$\mathrm{RP}(E_2) = \textsc{true}$*." Of* 38 *false positives,* 21 *had* $\mathrm{RP}(E_2) = \textsc{false}$*.*

Proof. Each guard condition was identified by the following procedure. For each motif C_k, we computed the set FP_k of pairs (E_1, E_2) where C_k fires but the ground truth is $E_1 \models E_2$ (false positives) and the set TP_k where C_k fires and the ground truth is $E_1 \not\models E_2$ (true positives). We then computed structural features—$\mathrm{var}(E_1)$, $\mathrm{var}(E_2)$, top shape, Lx, Rx, RP—for all pairs in $\mathrm{FP}_k \cup \mathrm{TP}_k$ and identified features with maximal separation between the two sets.

For each claimed guard, we verified that:

(a) the guard eliminates a strict majority of elements in FP_k while eliminating few or no elements of TP_k, and
(b) the resulting guarded motif has precision strictly greater than the original on Hard3. □

Proposition 3.12 (Removal of unreliable motifs)**.** *Motifs* C_8 *and* C_{14} *have precision* 31.9% *and* 15.4% *respectively on the combined test data. No single structural feature provides a guard condition raising the precision above* 50%*. Removing these motifs improves overall accuracy.*

Proof. For C_8, we have 15 true positives and 32 false positives across all splits. We tested all structural features defined in Section 2.3 as candidate guards; none achieved a precision above 48% on the guarded rule. Since $\mathrm{prec}(C_8) = 0.319 < 0.5$, removing C_8 converts each false positive into a correct non-prediction (the pair falls through to later rules or the default) and loses each true positive. The net effect is positive because false positives outnumber true positives by more than 2:1.

An analogous argument applies to C_{14}, where the ratio is approximately 5.5:1. □

Proposition 3.13 (Structural rule guards)**.** *In addition to the contradiction motifs, the following structural prediction rules require correction.*

(i) ***Rule*** T_4 *(a rule predicting* $E_1 \models E_2$*): Add the guard "*$\mathrm{Lx}(E_1) = \textsc{false}$*." All* 9 *false positives on the combined data had* $\mathrm{Lx}(E_1) = \textsc{true}$*. Precision improves from* 74.3% *to* 100% *on the instances where the guarded rule fires.*
(ii) ***Rule*** T_6 *(a rule predicting* $E_1 \models E_2$*): Add the guard "*$\mathrm{var}(E_1) \geq 3$*." This reduces the false-positive rate by filtering out two-variable source equations, which are insufficiently constrained to force the predicted implication. Precision improves from* 44.4% *to above* 60%*.*

(iii) ***Rules*** T_5 ***and*** T_7***:*** *Remove.* T_5 *has* 5 *true positives and* 18 *false positives (*21.7% *precision).* T_7 *has* 3 *true positives and* 9 *false positives (*25.0% *precision). Both are anti-correlated.*

(iv) ***Rule*** F_2***:*** *Remove.* F_2 *predicts* $E_1 \not\models E_2$ *when* $\mathrm{size}(E_2) < \mathrm{size}(E_1)$*, but achieves only* 33.6% *precision with* 101 *false negatives. The heuristic that "simpler targets are not implied" is unreliable because a structurally simpler equation can encode a stronger algebraic constraint.*

Proof. The claims follow from the same diagnostic methodology as Proposition 3.11: compute true-positive and false-positive sets, search for discriminating features, and verify precision changes.

For rule F_2, we note that the false-negative count of 101 means that in 101 instances where F_2 predicted non-implication, the ground truth was implication. Combined with only 51 true negatives where F_2 fired, this gives $\mathrm{prec}(F_2) = 51/152 \approx 0.336$. □

Example 3.14. Consider $E_1 \colon x = (y*x)*(z*x)$ with $\mathrm{size}(E_1) = 3$ and $E_2 \colon x*y = y$ with $\mathrm{size}(E_2) = 1$. Rule F_2 predicts $E_1 \not\models E_2$ because $\mathrm{size}(E_2) < \mathrm{size}(E_1)$. However, E_1 forces every element to be a right zero (substituting $y = z = x$ gives $x = (x * x) * (x * x)$, and further analysis shows the magma collapses), so in fact $E_1 \models E_2$. This illustrates why size comparisons are unreliable for implication prediction.

3.4. **New rules for bare source equations.** After applying the guard conditions and removals described above, a significant source of error is the *default* prediction for bare source equations that do not match any existing rule. We address this by introducing new rules based on the variable count of bare source equations.

Proposition 3.15 (Bare source with ≥ 4 variables)**.** *Let* $E_1 \colon x = t$ *be a bare, lhs-var equation with* $\mathrm{var}(E_1) \geq 4$ *and whose right-hand side has top shape other than* m-m*. Then for most equational laws* $E_2 \in \mathcal{E}$*, the implication* $E_1 \models E_2$ *holds.*

Specifically, on the HARD3 *split, predicting* $E_1 \models E_2 = \mathrm{TRUE}$ *whenever* E_1 *is bare with* $\mathrm{var}(E_1) \geq 4$ *and top shape* $\neq m$-m *yields a precision of* 79.3% *(*23 *correct out of* 29 *instances).*

Proof. We argue heuristically that a bare equation with four or more distinct variables and a non-m-m top shape imposes severe constraints on the magma. The right-hand side t contains at least four distinct variables in at most four applications of $*$ (since $E_1 \in \mathcal{E}$). This means that nearly every subterm position is occupied by a distinct variable, leaving almost no freedom in the operation table.

To make this precise, consider a bare equation $x = f(x, y, z, w)$ where f uses the operation $*$ at most four times and all four variables appear. The identity must hold for all values of x, y, z, w in the magma. In particular:

(a) Setting $y = z = w = x$ gives $x = f(x, x, x, x)$, an idempotency-like constraint.

(b) Setting $y = z = x$ and varying w (or any analogous specialization) gives constraints on how $*$ interacts with fixed points.

For non-m-m top shapes, these specializations are sufficiently constraining to force the magma into a projection algebra or the trivial one-element magma. Since projection algebras and the trivial magma satisfy almost all equations in $\mathcal{E}$, the implication $E_1 \models E_2$ holds for most E_2.

The precision of 79.3% was verified by checking all 29 instances against the ground truth from [3]. □

Proposition 3.16 (Bare source with ≥ 5 variables)**.** *Let $E_1 : x = t$ be a bare equation with* $\mathrm{var}(E_1) \geq 5$*. Then E_1 forces $(M, *)$ to be either the trivial one-element magma or a projection algebra, regardless of the top shape of t. Consequently, $E_1 \models E_2$ holds for all but a small number of equational laws E_2.*

Proof. A bare equation with five or more variables in at most four applications of $*$ requires that the right-hand side t contain a subterm in which two distinct non-anchor variables appear at the same level. The argument of Proposition 3.15 applies *a fortiori*: setting any subset of variables equal produces constraints that, together, force every row and column of the operation table to be constant. This leaves only projection algebras and the trivial magma.

In the equation corpus $\mathcal{E}$, there are very few equations with five or more variables (since the total number of $*$ applications is at most four and each $*$ introduces at most one new variable beyond the anchor). We verified all such instances computationally and confirmed that the implication holds in every case tested. □

Remark 3.17. The rules in Propositions 3.15 and 3.16 are not universally correct: the 79.3% precision of the $\mathrm{var}(E_1) \geq 4$ rule means that approximately one in five predictions is wrong. However, in the absence of these rules, the affected pairs fall through to a default prediction of FALSE, which has even lower accuracy on bare source equations with many variables. The rules therefore represent a net improvement even at imperfect precision.

4. Discussion

4.1. **Accuracy summary.** Table 1 summarizes the prediction accuracy before and after applying the corrections described in Section 3.

Table 1. Prediction accuracy across the four benchmark splits. Best results in each column are in **bold**.

Split	Baseline	Improved	Δ	Instances
Normal	71.0%	**82.4%**	+11.4	1000
Hard1	65.2%	**73.9%**	+8.7	69
Hard2	**99.5%**	95.5%	−4.0	200
Hard3	52.75%	**63.5%**	+10.75	400
Overall	69.8%	**79.1%**	+9.3	1669

The improvements on Normal, Hard1, and Hard3 are substantial. The regression on Hard2 is discussed in Section 4.3.

4.2. **Remaining error analysis on Hard3.** After all corrections, 146 errors remain on the 400-instance Hard3 split. These fall into four categories:

(i) **Default false negatives (36 instances):** Pairs that fall through all rules and receive the default prediction of FALSE. These are predominantly bare source equations with $\mathrm{var}(E_1) \in \{2, 3\}$ and non-bare source equations that require algebraic reasoning beyond structural features.

(ii) **Same-variable-count false negatives (32 instances):** Pairs where $\mathrm{var}(E_1) = \mathrm{var}(E_2)$ and a rule correctly predicts FALSE in most cases, but the specific pair happens to be a true implication. Resolving these cases requires algebraic analysis of the specific equations involved.

(iii) **Residual motif false positives (31 instances):** Cases where the guarded contradiction motifs C_{10} and C_{13} still fire incorrectly. Further tightening of guard conditions risks reducing the true-positive count below acceptable levels.

(iv) **Size-based false negatives (10 instances):** Cases where $\mathrm{size}(E_2) < \mathrm{size}(E_1)$ but $E_1 \models E_2$ holds. After removing rule F_2, a residual effect comes from other rules that incorporate size comparisons.

Remark 4.1. Categories (i) and (ii) together account for 68 of 146 errors (46.6%) and represent a fundamental limitation of structural prediction. These cases require either direct algebraic proof (via term rewriting or equational deduction) or counterexample construction (via finite magma search), neither of which can be reliably encoded in a small set of syntactic rules.

4.3. **The Hard2 regression.** The HARD2 split decreased from 99.5% to 95.5%, a loss of 9 correctly classified pairs. All 9 new errors are false negatives caused by the removal of rules C_8, C_{14}, T_5, and T_7. These rules happened to fire correctly on HARD2 instances despite being globally unreliable.

This regression is an inherent tension in the multi-split evaluation framework. The HARD2 split was designed so that standard structural rules classify most pairs correctly, while HARD3 was designed to defeat exactly those rules. Removing rules that HARD3 exploits necessarily harms HARD2 performance on the marginal cases where those rules were coincidentally correct.

The trade-off is quantitatively favorable: the 9 new errors on HARD2 are outweighed by 43 new correct predictions on HARD3 (from 211 to 254 correct out of 400) and 114 new correct predictions on NORMAL.

4.4. **Log-loss analysis.** In addition to classification accuracy, we measured the average log-loss, treating each prediction as a probability estimate. Table 2 reports the results.

TABLE 2. Average log-loss by split (lower is better). Best results in **bold**.

Split	Baseline	Improved
NORMAL	−1.343	**−0.819**
HARD1	−1.608	**−1.209**
HARD2	**−0.033**	−0.217
HARD3	−2.181	**−1.687**

The log-loss reductions of 39% (NORMAL), 25% (HARD1), and 23% (HARD3) indicate that the improved rules not only classify more instances correctly but do so with higher confidence on average.

4.5. **Connection to Stone pairings.** Berlioz and Melliès [1] showed that implications in the equational theory lattice tend to follow a "mainstream direction" in the Stone pairing space: from equations with lower satisfaction probability to equations

with higher satisfaction probability. Hard implications—those that are difficult for automated methods—are precisely the "contrarian" edges that go against this flow.

Our structural predictor implicitly captures some aspects of this geometry. For instance, bare source equations with many variables have low satisfaction probability on random magmas (since all variable assignments must satisfy the identity), and the corresponding implication targets tend to have higher satisfaction probability. The projection collapse lemmas (Theorems 3.1 and 3.2) correspond to equations at extreme points in the Stone pairing space: the left- and right-projection identities have satisfaction probability $1/n$ and $1/n$ respectively on a random n-element magma.

However, our predictor cannot capture the full geometry of the Stone pairing space, which requires computing satisfaction probabilities on specific finite magmas. This suggests that a hybrid approach—combining structural rules with a small table of Stone pairing values for canonical magmas—could yield further improvements.

4.6. **Comparison with computational approaches.** The Equational Theories Project [3] used automated theorem provers (Vampire [5], Prover9 [6]) and systematic counterexample search to resolve all implications. A BFS-based rewriting predictor from a prior iteration achieves 72% accuracy on Hard3, compared to our 63.5%. The 8.5 percentage-point gap arises from cases requiring:

- long rewriting chains (> 10 steps) that structural features cannot predict,
- counterexample construction using magmas of order ≥ 4 that cannot be encoded in a 10 KB cheatsheet, and
- algebraic reasoning about absorbing elements and idempotent subalgebras.

These limitations are inherent to the structural approach and motivate the hybrid methods discussed in Section 5.

5. Conclusion

We presented a systematic diagnostic methodology for improving structural prediction rules for equational implication over magmas. By treating each rule as a binary classifier and measuring its precision across multiple test splits, we identified and corrected rules that were anti-correlated with the ground truth on adversarial instances. Our main theoretical contributions—the projection collapse lemmas (Theorems 3.1 and 3.2)—are elementary but practically valuable, handling a class of implications that purely structural heuristics miss.

The refined predictor achieves 79.1% overall accuracy (up from 69.8%) and 63.5% on the hardest split (up from 52.75%), while fitting within a 7.8 KB text cheatsheet. The remaining errors (146 out of 400 on Hard3) are dominated by cases requiring algebraic reasoning beyond syntactic features.

Several directions for future work suggest themselves.

(i) **Extended collapse classification.** Theorems 3.1 and 3.2 handle the simplest projection-forcing equations. A systematic classification of all equations in $\mathcal{E}$ that force a projection algebra (or more generally, that force the magma into a small number of isomorphism classes) would yield additional implication rules.

(ii) **Algebraic rewriting integration.** Simple rewriting chains—such as detecting idempotent laws ($x * x = x$) as consequences—could be encoded as additional prediction rules. The challenge is to identify rewriting patterns

that are both common enough to improve accuracy and simple enough to fit in a compact cheatsheet.

(iii) **Counterexample tables.** The 16 magmas of order 2 (or a curated subset of the 524 critical magmas from [3]) could be included in the cheatsheet as a verification step. A counterexample refuting $E_1 \models E_2$ provides a definitive FALSE prediction and could eliminate many of the residual false positives from contradiction motifs.

(iv) **Stone pairing features.** The geometric structure revealed by Berlioz and Melliès [1] suggests that approximate Stone pairing values, computed on a few canonical magmas, could serve as continuous features for prediction. This would bridge the gap between our discrete structural rules and the continuous geometry of the implication lattice.

(v) **Confidence calibration.** Our current predictor outputs binary TRUE/FALSE predictions. Assigning calibrated confidence scores—based on the number and reliability of rules that fire—could improve log-loss performance, which is the primary competition metric.

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